

Knowledge Graphs in Production

1. Entity Resolution at Scale

Production knowledge graphs ingest records from dozens of upstream systems. Entity resolution merges duplicate nodes that describe the same real-world object. In our deployment the resolver runs as a streaming job: blocking keys candidate pairs, a similarity model scores them, and merges above 0.92 are applied automatically while the 0.75-0.92 band is queued for human review.

2. Schema Evolution

Schemas change weekly. Backward-compatible changes (adding optional properties, new node types) ship without migration. Breaking changes follow a two-phase rollout: a dual-write period of at least two weeks, then a cutover once read paths are migrated.

3. Operational Lessons

Graph health is monitored through three signals: dangling-edge rate, entity-merge precision sampled weekly, and query latency at the 99th percentile. Incremental materialization keeps the serving store fresh within five minutes of an upstream write; full rebuilds run monthly and are validated against the incremental store before swap. The most expensive incident traced back to a silent schema drift between writer and reader versions, which is why dual-write windows are now mandatory.